Signals and Systems Lecture-1: Introduction

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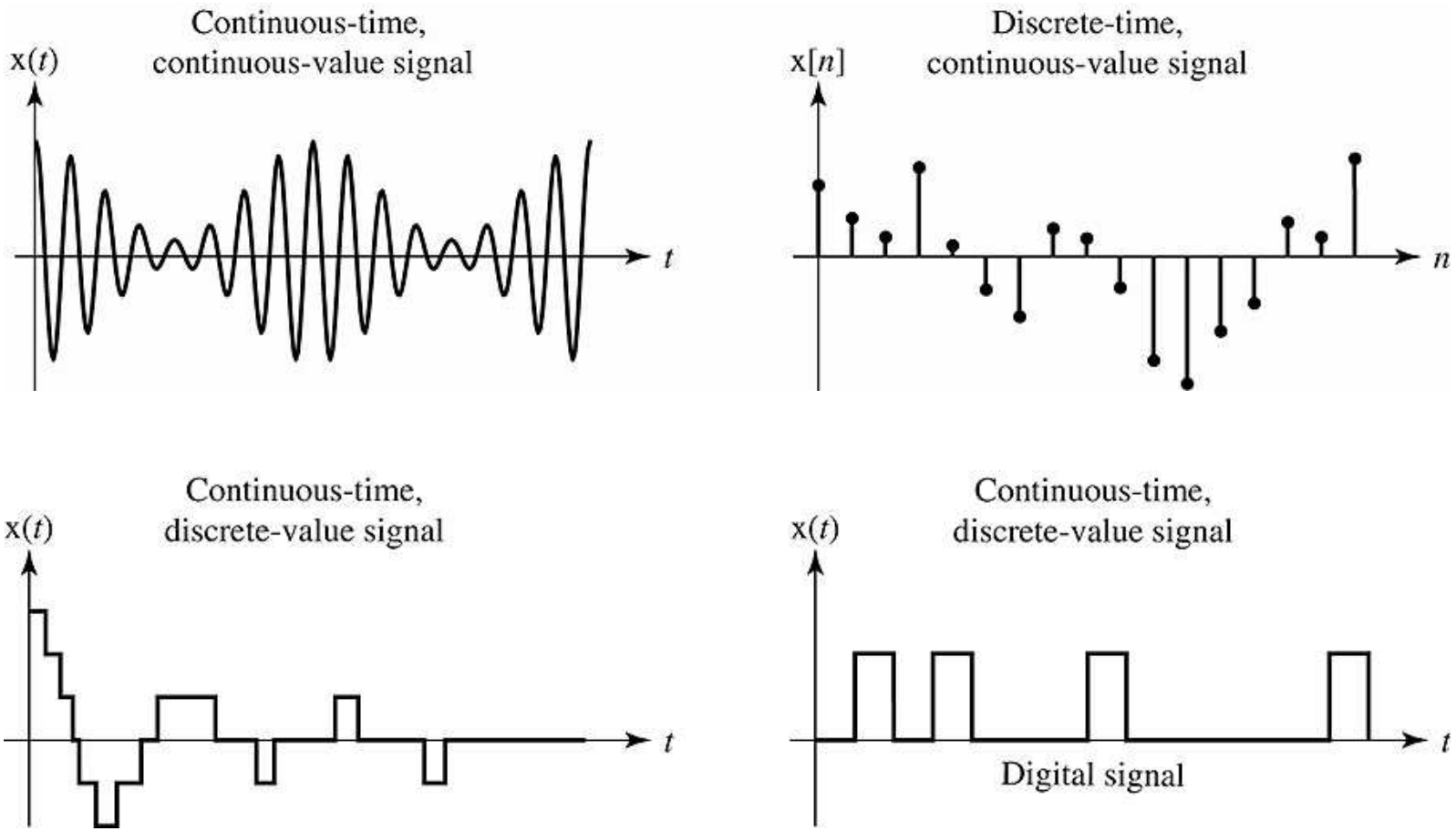
January 16, 2023

Introduction (1)

- A Signal corresponds to any time-varying physical phenomenon which is intended to convey information.
  - Human voice.
  - Voltages on telephone wires.
  - Electromagnetic waves from radio and television transmitters.
- Noise corresponds to a time-varying random signal that does not carry useful information. It is almost always considered as undesirable.
- Systems are used to process or to operate on signals.

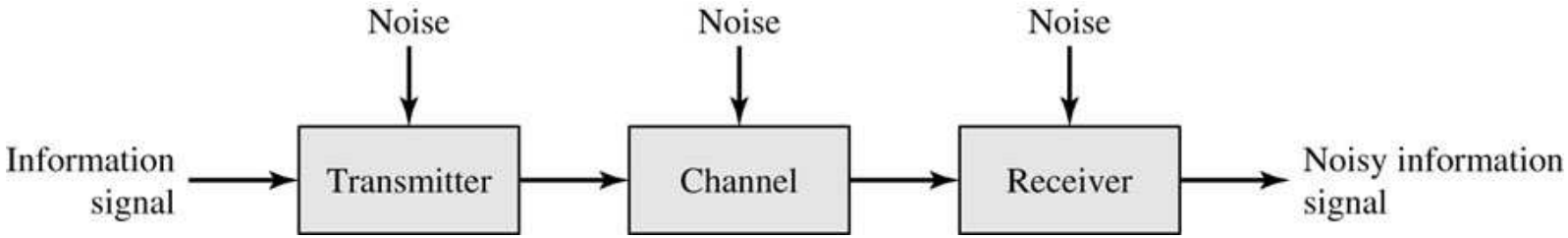
Introduction (2)

- Signals can have:
  - A Continuous-Time (CT) or a Discrete-Time (DT).
  - A Continuous-Amplitude (CA) or a Discrete-Amplitude (DA).

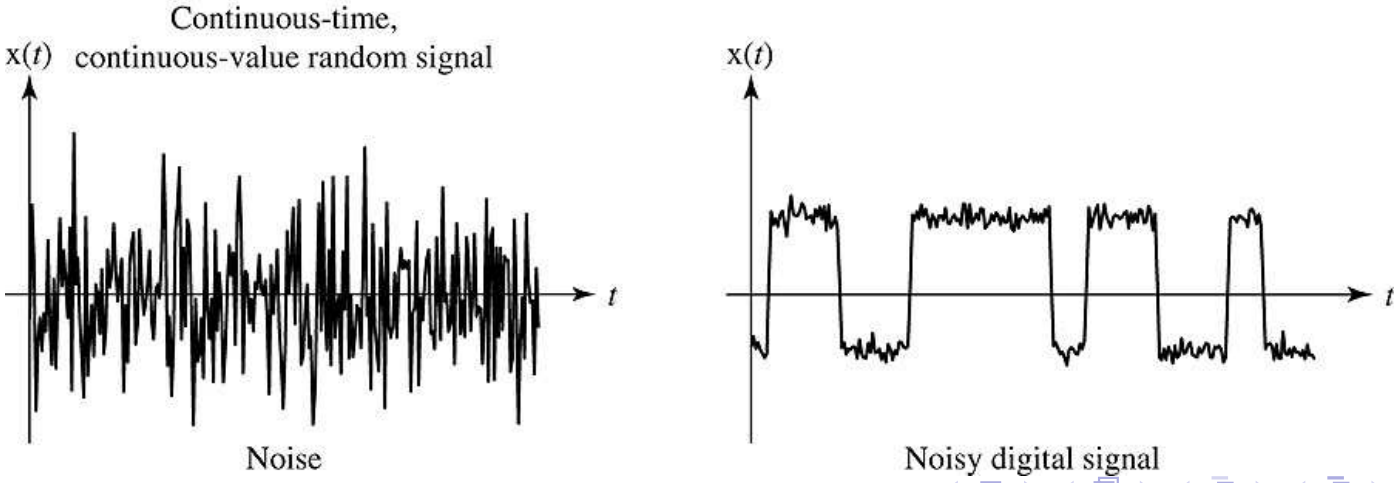


Introduction (3)

- In communication systems, noise results from:
  - The electronic components at the transmitter and the receiver.
  - The communication channel, for example, interference from other users.



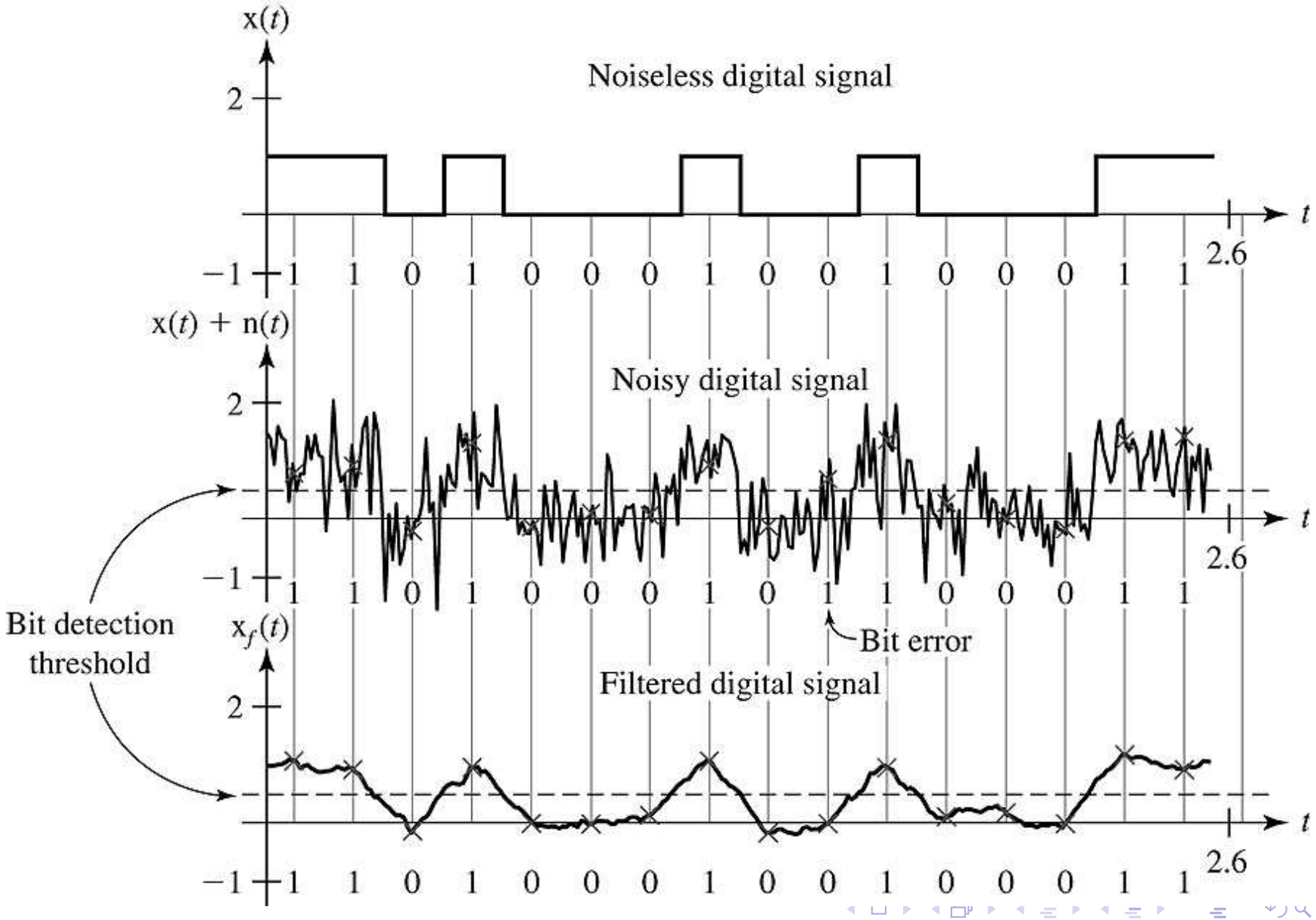
- The next figure shows an example of noise and a noisy digital signal:



Introduction (4)

- There has been a great interest in the design of digital signals and digital systems.
- Digital signals have the advantage of a better immunity to noise.
  - In binary communications, the bits can be detected very cleanly until the noise gets very large.
  - The signal value at the middle of each bit duration is compared to a threshold:
    - If this value is above the threshold, a 1 is detected.
    - If this value is below the threshold, a 0 is detected.

Introduction (5)

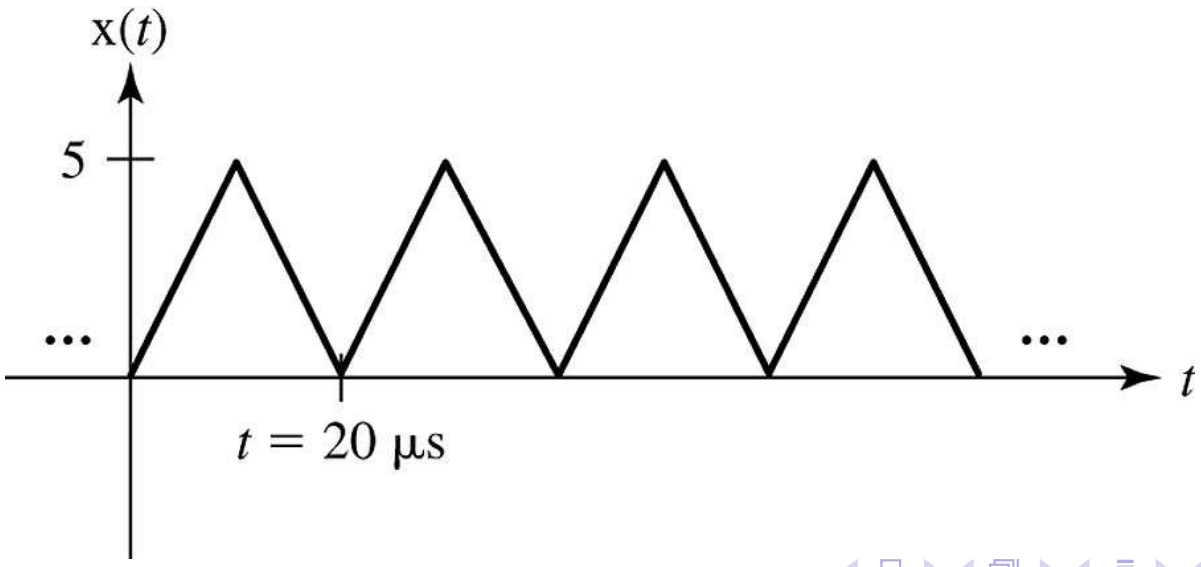


Periodic Signals (1)

- A periodic signal with period T_0 repeats itself every T_0 seconds.
- In other words, a periodic continuous-time signal $x(t)$ verifies the following relation:

$$x(t) = x(t + nT_0) \ \forall \ n \in \mathbb{Z}$$

  Consequently, the signal is invariant under the transformation $t \rightarrow t + nT_0$.
- For example, the next signal is periodic with $T_0 = 20\ \mu$s.



Periodic Signals (2)

- Consider the CT sinusoidal signal:

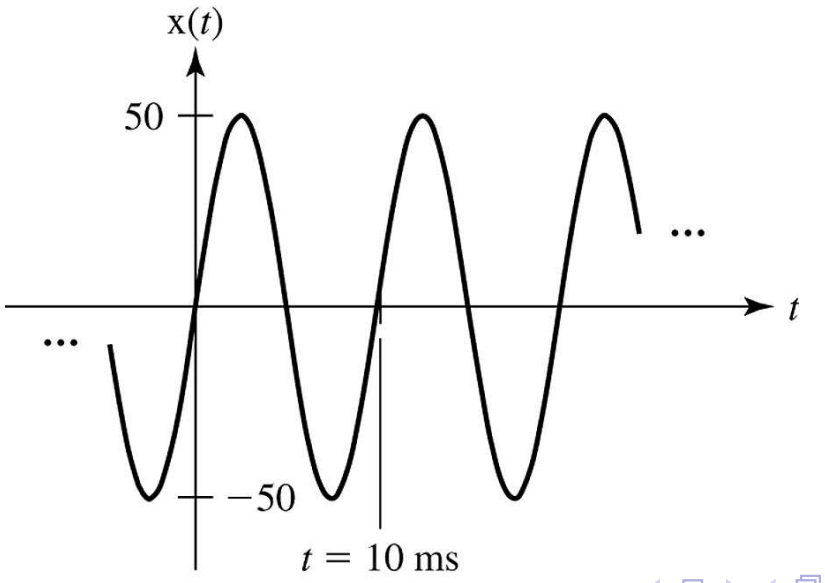
$$x(t) = A\sin(2\pi f_0 t + \theta)$$

This signal is periodic with a period $T_0 = \frac{1}{f_0}$ where f_0 corresponds to the fundamental frequency of $x(t)$.

- For example, the next figure shows the signal:

$$x(t) = 50\sin(200\pi t)$$

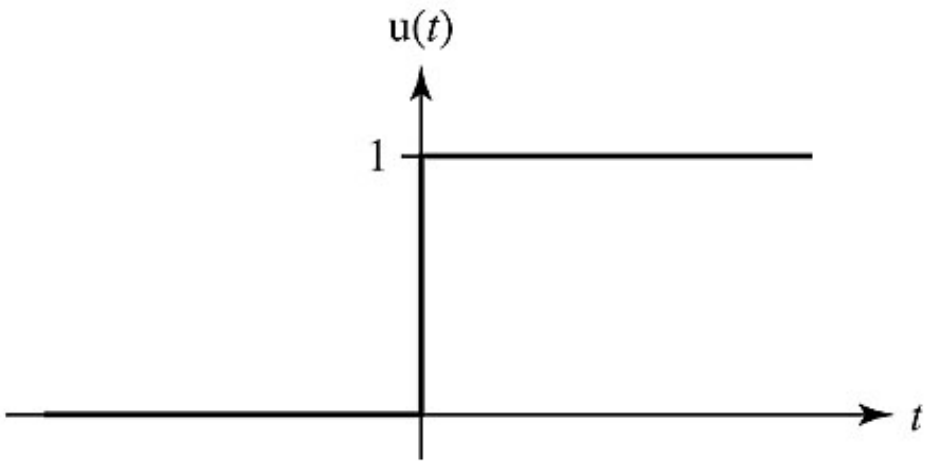
that has a frequency $f_0 = 100$ Hz and a period $T_0 = 10$ ms.



Some Common CT signals (1)

- The Unit Step Function is defined by:

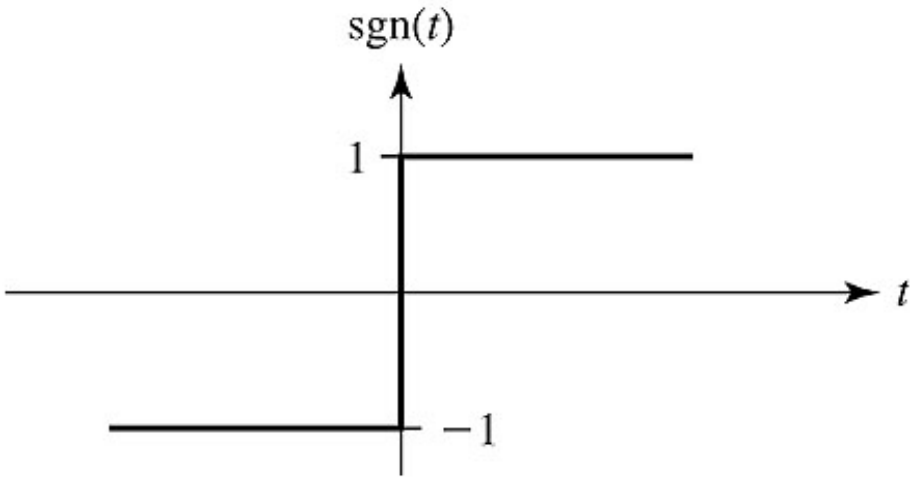
$$u(t) = \begin{cases} 1, & t \geq 0; \\ 0, & t < 0. \end{cases}$$



Some Common CT signals (2)

- The Signum Function is defined by:

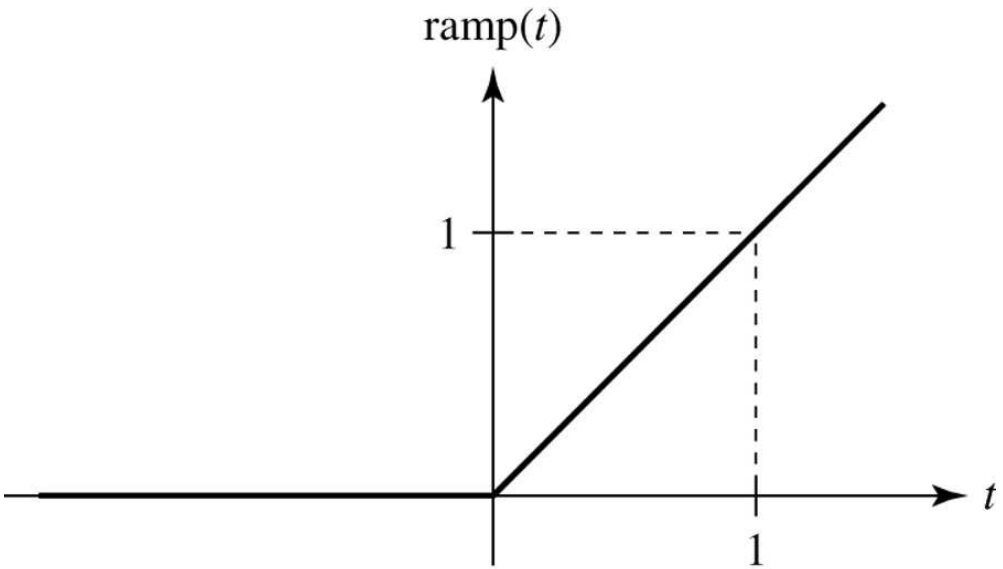
$$\mathrm{sgn}(t) = \begin{cases} 1, & t \geq 0; \\ -1, & t < 0. \end{cases}$$
$$= 2u(t) - 1$$



Some Common CT signals (3)

- The Unit Ramp Function is defined by:

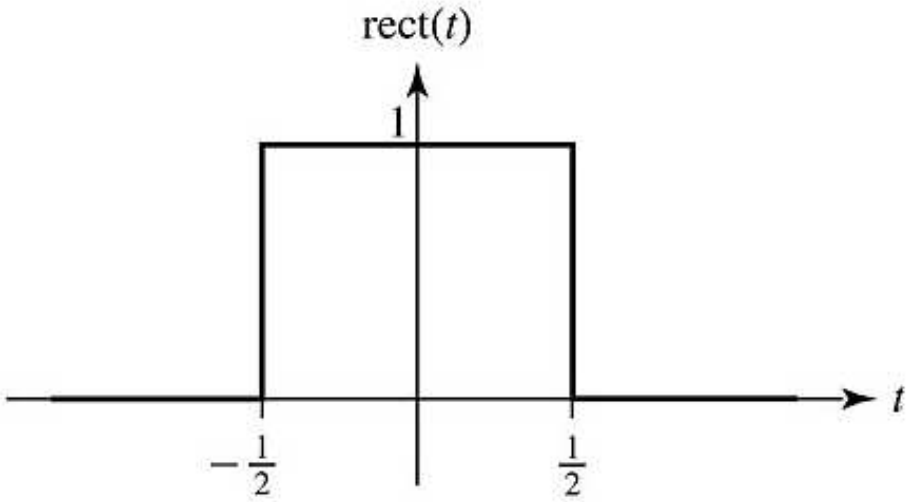
$$\text{ramp}(t) = \begin{cases} t, & t \geq 0; \\ 0, & t < 0. \end{cases}$$
$$= tu(t)$$



Some Common CT signals (4)

- The Unit Rectangle Function is defined by:

$$\mathrm{rect}(t) = \begin{cases} 1, & |t| \leq \frac{1}{2}; \\ 0, & |t| > \frac{1}{2}. \end{cases}$$

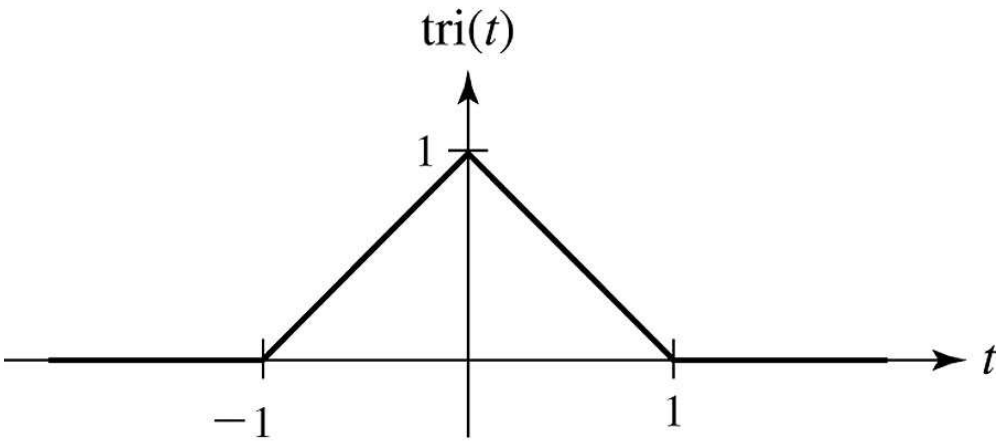


Some Common CT signals (5)

- The Unit Triangle Function is defined by:

$$\mathrm{tri}(t) = \begin{cases} 1 - |t|, & |t| \leq 1; \\ 0, & |t| > 1. \end{cases}$$

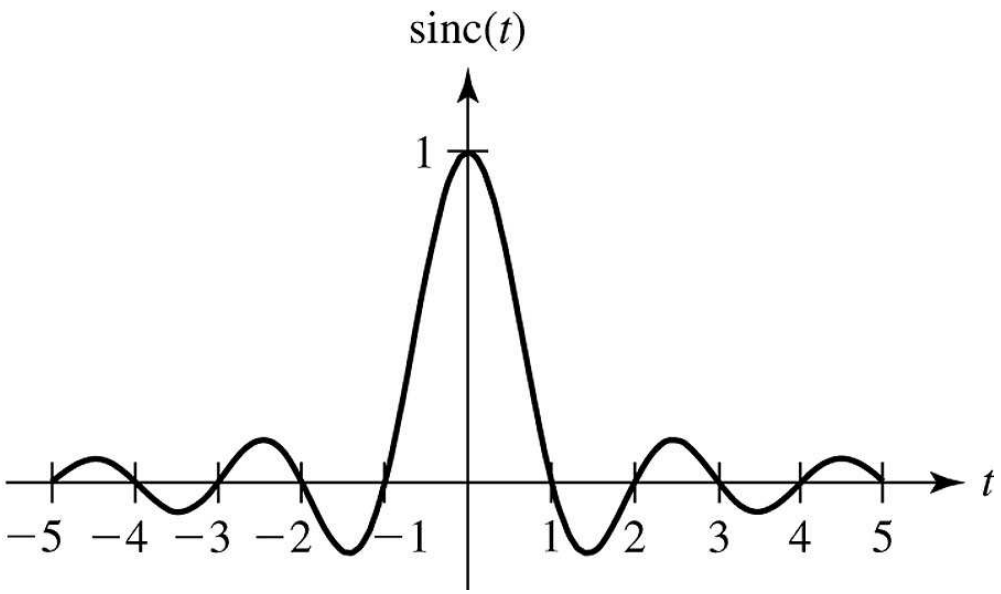
$$= \begin{cases} 1 + t, & -1 \leq t \leq 0; \\ 1 - t, & 0 \leq t \leq 1; \\ 0, & |t| > 1. \end{cases}$$



Some Common CT signals (6)

- The Unit Sinc Function is defined by:

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$



- Note that:

$$\text{sinc}(0) = 1$$
$$\text{sinc}(t) = 0 \;\; \text{for} \;\; t \in \mathbb{Z}^*$$